

Section 1: SAP Production Planning (PP) Architecture

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

SAP PP aligns manufacturing capabilities with customer demand. It covers demand management, sales and operations planning (S&OP), Material Requirements Planning (MRP), capacity planning, and shop floor order execution.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with SAP Production Planning (PP) Architecture include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within SAP Production Planning (PP) Architecture updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 2: Bills of Material (BOM) - CS01

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

A Bill of Materials (CS01) is a structured list of components, raw materials, and quantities needed to manufacture a product. BOMs can be single-level or multi-level, and are assigned to specific plants and usages.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Bills of Material (BOM) - CS01 include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Bills of Material (BOM) - CS01 updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 3: Work Centers (CR01)

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Work centers represent production locations, machines, or labor groups where manufacturing activities take place. Key data includes available capacity, formulas for scheduling and costing, and cost center assignments.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Work Centers (CR01) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Work Centers (CR01) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 4: Routings and Operations (CA01)

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

A Routing (CA01) defines the sequence of manufacturing operations required to produce a material. Each operation specifies the work center, standard operational times (setup, machine, labor), and component allocations.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Routings and Operations (CA01) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Routings and Operations (CA01) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 5: Demand Management and Planned Independent Requirements

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Planned Independent Requirements (PIRs, MD61) represent forecasted production quantities. Demand management strategies determine whether production is Make-to-Stock (MTS, strategy 10/40) or Make-to-Order (MTO, strategy 20/50).

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Demand Management and Planned Independent Requirements include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Demand Management and Planned Independent Requirements updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 6: Material Requirements Planning (MRP Live)

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

MRP compares requirements against warehouse inventory and planned receipts to generate planned orders and purchase requisitions. In S/4HANA, MRP Live runs in-memory directly on HANA, executing up to 10x faster than traditional ERP MRP.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Material Requirements Planning (MRP Live) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Material Requirements Planning (MRP Live) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 7: Production Order Lifecycle

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

The production order lifecycle encompasses: Order Creation (CO01), Order Release (CO02), Material Staging, Goods Issue of components (261), Operation Confirmation (CO11N), Goods Receipt of finished item (101), and Order Technical Completion (TECO).

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Production Order Lifecycle include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Production Order Lifecycle updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 8: Material Staging and Goods Issue (MIGO / 261)

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Material staging brings raw materials from warehouse bins to the shop floor. Goods Issue with movement type 261 consumes components against the production order, updating actual material costs in the order cost collector.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Material Staging and Goods Issue (MIGO / 261) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Material Staging and Goods Issue (MIGO / 261) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 9: Production Confirmation (CO11N)

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Production confirmation records completed operational quantities, scrap quantities, and actual machine/labor times. Confirmations update order status, reduce capacity requirements, and calculate activity costs.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Production Confirmation (CO11N) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Production Confirmation (CO11N) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 10: Goods Receipt from Production (MIGO / 101)

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Completed finished products are received into inventory with movement type 101. This updates stock in warehouse storage locations and credits the production order at standard cost.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Goods Receipt from Production (MIGO / 101) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Goods Receipt from Production (MIGO / 101) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 11: Capacity Requirements Planning (CRP)

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1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Capacity planning analyzes workload against available work center capacity over time buckets. Capacity leveling tools resolve overloads by adjusting operation start dates or shifting work to alternative machines.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Capacity Requirements Planning (CRP) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Capacity Requirements Planning (CRP) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 12: Repetitive Manufacturing (PP-REM)

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Repetitive manufacturing is used for continuous, high-volume production lines. Production occurs against run schedules rather than discrete orders, and backflushing (MFBB) posts simultaneous component consumption and goods receipt.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Repetitive Manufacturing (PP-REM) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Repetitive Manufacturing (PP-REM) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 13: Subcontracting in Production Orders

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

External operations in routings subcontract specific manufacturing steps (e.g. heat treatment, painting) to outside vendors, generating subcontracting purchase requisitions automatically.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Subcontracting in Production Orders include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Subcontracting in Production Orders updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 14: Scrap Management in Production

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Component scrap, assembly scrap, and operation scrap account for anticipated and actual manufacturing losses during production, ensuring MRP orders sufficient raw materials to yield required finished quantities.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Scrap Management in Production include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Scrap Management in Production updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 15: Batch Management Integration

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Batch management assigns unique batch numbers to raw materials and finished products, enabling complete forward and backward traceability throughout the manufacturing supply chain.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Batch Management Integration include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Batch Management Integration updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 16: Engineering Change Management (ECM)

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Engineering Change Management (CC01) manages modifications to BOMs, routings, and product designs with audit logging and effective date controls to maintain regulatory compliance.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Engineering Change Management (ECM) include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Engineering Change Management (ECM) updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 17: Production Order Settlement and Variance

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Production orders are settled at month-end (KO88). The difference between actual costs debited and standard costs credited is posted to price difference accounts in general ledger accounting.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Production Order Settlement and Variance include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Production Order Settlement and Variance updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 18: Shop Floor Control and Dispatching

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Shop floor control coordinates order sequencing and dispatching at individual work centers to minimize setup times and optimize machine utilization.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Shop Floor Control and Dispatching include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Shop Floor Control and Dispatching updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 19: Integration with QM and PM

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

PP integrates with Quality Management (QM) for in-process inspection lots (type 03) and with Plant Maintenance (PM) to schedule machine preventative maintenance without disrupting production.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Integration with QM and PM include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Integration with QM and PM updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.

Section 20: Common PP Operational Errors

Module: SAP PP | Document: SAP_PP_Production_Planning_Reference.pdf

1.0 TECHNICAL SPECIFICATION AND ARCHITECTURE

Typical PP issues include missing component allocations in routings, scheduling errors due to missing work center formulas, and confirmation blocks. Resolutions involve checking CA02, CR02, and COFC error logs.

2.0 OPERATIONAL AND CONFIGURATION DETAILS

In an enterprise production environment running SAP PP, this functionality is configured and maintained according to SAP Best Practice methodologies. System administrators and functional consultants must ensure that all prerequisites, master data records, and organizational assignments are established prior to execution.

Key transaction codes, database tables, and verification checkpoints associated with Common PP Operational Errors include:

- Primary Control Points: Ensure organizational authorization and period readiness.
- Data Integrity: Automated cross-table validation ensures zero orphaned records.
- Audit Compliance: System logs, change documents, and audit trails capture all transactional postings.

3.0 INTEGRATION TOUCHPOINTS

Seamless integration across logistics and finance is the hallmark of the SAP ERP ecosystem. Any transactional event within Common PP Operational Errors updates general ledger postings, operational statuses, and downstream planning tools in real-time, eliminating asynchronous reconciliation steps.